

Viral Mutation Classification using DNABERT-2 and LSTM

D Vamsi Krishna

CSE - Data Science

R.V.R & J.C.C.E

Guntur, Andhra Pradesh

vamsi.vk1248@gmail.com

Mrs. K Aravinda

CSE - Data Science

R.V.R & J.C.C.E

Guntur, Andhra Pradesh

aravinda@vrjc.ac.in

G Samba Siva Rao

CSE - Data Science

R.V.R & J.C.C.E

Guntur, Andhra Pradesh

sambasivaragorantla1@gmail.com

D Ajay Kumar Reddy

CSE - Data Science

R.V.R & J.C.C.E

Guntur, Andhra Pradesh

doddaajaykumarreddy@gmail.com

Abstract— New developments in Natural Language Processing (NLP) and transfer learning have provided new opportunities in the analysis of genomic sequences. This paper is based on exploring the advantages of DNABERT-2 and a hybrid DNABERT-LSTM model on mutation classification in viral genomes. The dataset that was employed in the study was retrieved at the NCBI Viruses repository and has several viral genome sequences characterized as wildtype or mutated. DNABERT-2 is based on k-mer tokenization and transformer attention to build relationships in context in DNA sequences and Long Short-Term Memory (LSTM) to build long-range dependencies in sequential data. The suggested hybrid model is a combination of DNABERT-2 with regard to its contextual representation abilities and LSTM with regard to its sequential learning capabilities to enhance classification performance. Indeed, experimental assessment was performed based on such indicators as accuracy and F1-score on various viral datasets, Human Virus, Influenza Virus, and SARS-CoV-2. The findings indicate that the hybrid DNABERT-LSTM model has a higher classification accuracy and generalization than the conventional ones. The paper shows the capability of NLP-based deep learning models to perform scalable mutations analysis of the genome and gives a template of how AI-driven genomic studies can be conducted in the future.

Keywords— DNABERT, Mutation Classification, Genomic Sequencing, Natural Language Processing, Transformer Models, Bidirectional LSTM, Transfer Learning, Viral Genome Analysis, Deep Learning, Bioinformatics.

I. INTRODUCTION

The recent swift expansion in sequencing methods has created vast quantities of genomic information and created the possibilities of new insights into biological mechanisms and the pathology of diseases. Medical diagnostics, evolutionary biology, and virology are the areas of interest that need genomic sequence analysis. Identification of mutations is one of the most important activities in genomic analysis and is a sequence alteration that may influence expression of any gene, protein activity and disease progression. In viral genome studies, mutations can modify the transmissibility, virulence and

resistance to anti-viral therapy, thus accurate detection of mutations is particularly important.

The classical approaches of bioinformatics principally use sequence comparison and reference based mapping as the techniques based on alignments. Nevertheless, they do not tend to address large genomic datasets and complicated mutation patterns. They are computationally costly and they might not be able to model long-range interactions between DNA sequences.

There are new methods of genomic analysis, which have been introduced by recent developments in deep learning and Natural Language Processing (NLP). The DNA sequences may be regarded as a biological language, in which the nucleotides are considered to be a character, and the k-mer sequences relate to the words. Transformer-based models like DNABERT are learned to provide contextual representations of DNA sequences by attention mechanisms.

Nonetheless, the transformer models do not necessarily reflect long-range sequential dependencies. The latter can be used to overcome this limitation by integrating them with recurrent neural networks like Long Short-Term Memory (LSTM).

The paper examines the mutation classification of viral genomes using a hybrid deep learning model consisting of DNABERT-2 and LSTM. DNABERT-2 derives contextual information of the DNA sequences and LSTM derives sequential information. The model is tested with the viral genomic data sets of NCBI Virus database including Human Virus, Influenza Virus and SARS-CoV-2. Findings indicate that hybrid model enhances better performance in mutation classification and offers a scalable system in studying the genomic sequence.

II. PROBLEM STATEMENT

The classification of genomic mutation is important in the process of comprehending viral evolution, disease progression, and the creation of effective medicine. Nevertheless, the classical bioinformatics tools that are used in detecting mutations are based on sequence alignment and rule-based analyses. These approaches are generally difficult to scale to the growing amount of genomic data and its complexity, particularly in the analysis of viral genomes that are rapidly evolving. The reliance on reference genomes is also a significant limitation of alignment-based methods, potentially preventing the detection of new or unusual mutations in viruses that are newly emerged.

The next problem in analyzing genomic mutations is that there are complicated patterns and long-range interactions in the DNA sequences. Traditional machine learning models typically demand some manual feature engineering and may not be able to encode the contextual dependencies among nucleotides in large genomes. Also, there are few labelled genomic datasets available, so it is hard to make highly accurate models using conventional supervised learning methods.

As the techniques of deep learning and Natural Language Processing (NLP) have become available, genomic sequences may be processed like language-like information, and with help of these methods, models can learn meaningful patterns directly out of raw sequences. Transformer-based models like DNABERT have however been primarily formulated to capture contextual dependencies, and may not completely capture sequential dependencies over long genomic regions.

Consequently, there exists the need to have a hybrid deep learning system that can be useful in capturing contextual and sequential properties of genomic sequences to enhance the accuracy of mutations classification. The proposed study attempts to solve the issue by developing a hybrid model of DNABERT-2 and LSTM that involves transformer-based embeddings and sequential learning to correctly classify mutation in viral genomic sequences.

III. RELATED WORK

The last developments in deep learning and Natural Language Processing (NLP) enhanced the analysis of genomic sequences. Models based on transformers, recurrent neural networks and hybrid-based models have been investigated to improve the detection of mutations and genomic classification.

Transformer models have also received interest in genomics to learn contextual relationships in biological sequences. Adapted to the BERT model, DNABERT considers the DNA sequences to be a language. DNABERT acquires contextual representation of nucleotide sequences by use of k-mer tokenization and attention to recognize functional genomic elements. Research indicates that DNABERT is superior to conventional machine learning algorithms in activities like prediction of promoters and classification of mutations.

A better variant DNABERT-2 was created in order to process multiple species genomic information and large biological data more effectively. DNABERT-2 is better optimized in transformer architecture and training plans, which result in greater generalization and accuracy in genomic classification tasks.

Hybrid frameworks that consist of transformer architecture with other neural networks have also been investigated by researchers. To illustrate, CNNs with DNABERT embedding representations learn local sequence motifs whereas the Long Short-Term Memory (LSTM) networks learn sequence-based and long-range interactions in DNA sequences.

The recent works combine DNABERT embeddings with LSTM networks, transformer models are used to produce contextual features, and LSTM is used to recreate sequential features. Although these developments have been made, there are some difficulties related to limited labeled datasets and the inability to generalize the results to other viral genomes.

In order to overcome these problems, this study will suggest a hybrid DNABERT-2 and LSTM approach to the task of mutation classification in viral genomes. The given strategy is expected to enhance the performance of the classification and generalization because it will be possible to conduct contextual learning and sequential learning on a variety of viral genomics datasets.

IV. OBJECTIVES

The principal goal of the study is to come up with a viable deep learning system in the classification of mutations in viral genomic sequences using Natural Language Processing programs.

The objectives of the study are specific and are:

1. To create a hybrid deep learning network with DNABERT-2 and Long Short-Term Memory (LSTM) that will be used to classify genomic mutations.
2. To extract contextual features of DNA sequences with the help of k-mer tokenization and transformer-based embeddings.
3. To learn long-range sequential interactions on genomic data with LSTM networks.
4. To test the suggested model on viral genomic data, including Human Virus, Influenza Virus, and SARS-CoV-2 of NCBI Viruses database.
5. To enhance the accuracy and reliability of mutation detection during the genomic sequence analysis.

V. METHODOLOGY

This paper provides a hybrid deep learning framework that combines the transformer-based model contextual learning and sequential modeling of recurrent neural networks to classify genomic mutations. The proposed system will involve the usage of DNABERT-2 to draw out features and Long Short-Term Memory (LSTM) networks in sequential modeling. The system is intended to give an appropriate classification of the genomic sequences as being in the wild type or mutated form by the manner in which it captures the contextual and long range dependencies in the DNA sequences.

The overall workflow of the proposed system consists of several steps that comprise the receiving of the dataset, the preprocessing stage, the tokenization of k-mers, the creation of embedding with the assistance of the transformer-based system, the sequential modelling with the LSTM, and final mutation classification.

A. Dataset Collection

The data source in the present study was the NCBI Viruses database that houses genomic sequences of various viral families. This database contains curated viral genome sequences and also contains mutation related information that are useful in genomic classification activities.

The dataset is marked according to genomic sequences, namely, wildtype and mutated. Wildtype class is a normal sequence of genome with no mutations and the mutated class is a sequence that has some genetic variation. The labels allow the model to do binary classification in detecting mutations.

These viruses are human viruses, influenza viruses, and

coronaviruses of the SARS-CoV-2 virus, with their viral genomes being included in the dataset. Simultaneously, the proposed model can be used to test its accuracy with various genomic structures and mutation patterns by using various viral datasets.

The information on the genomic sequences is set and prepared before training the model to make it compatible with the deep learning framework. The proposed methodology takes these sequences as input to preprocess, tokenize and extract features.

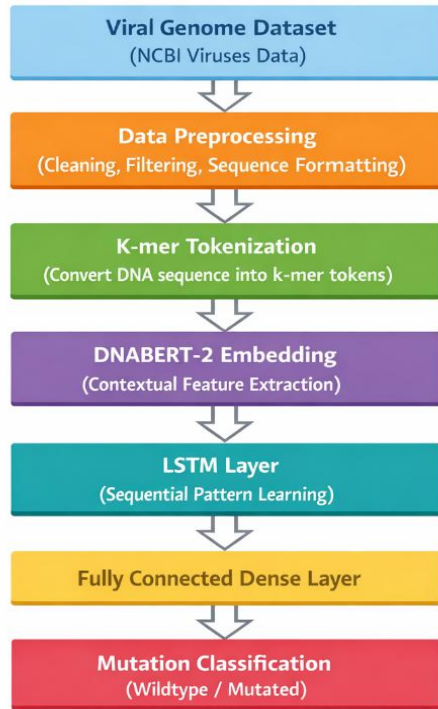


Fig 1: Methodology Overview

B. Data Preprocessing

The genomic sequences will have to be converted into an appropriate format that will be consumed by the deep learning models. Preprocessing stage involves numerous processes that will be used to test the data consistency and optimum model performance.

Preprocessing steps entail:

1. Sequence Cleaning:

Elimination of unfinished or corrupted data of the dataset.

2. Sequence Normalization:

Guaranteeing that all the genomical sequences (A, T, G, C) do not have invalid nucleotide characters.

3. Sequence Segmentation:

The long sequences of genome are broken down to smaller sequences in order to ensure that the model is fed with the same numbers of inputs.

Such measures can be used to make the training process more efficient and eliminate the noise factor on the model.

C. K-mer Tokenization

In order to metrically process genomic sequences with Natural Language Processing (NLP) technology, one will be required to encode the DNA strings into a tokenized form. This step is said to be k-mer tokenization.

A k-mer refers to a DNA sequence that is a fragment of k sequence. With $k = 6$, the DNA sequence is broken down to tokens with 6 nucleotides that overlap. The tokens also

enable the model to obtain local sequence patterns and biological motifs inherent in genomic data.

Example:

DNA sequence:

ATGCTGATCG

We taken k value as 6, then sequence in 6-mer tokens:

ATGCTG

TGCTGA

GCTGAT

CTGATC

The k-mer tokenization algorithm allows transformer based models to operate on genomic sequences just as they do with words in natural language. The model is able to learn valuable contextual relationships between patterns of nucleotides and enhance representation of genomic sequences by transforming the DNA sequences into tokens.

D. DNABERT-2 Feature Extraction

DNABERT-2 is a language model that is transformer-based and that was specifically modeled in the analysis of genomic sequences. It is a better alternative of the DNABERT model and is based on the Bidirectional Encoder Representations of Transformers (BERT) architecture. DNABERT-2 views the DNA sequences as a language and encodes the nucleotide sequences as k-mer tokens that may be learned by the model to establish contextual relations between the nucleotides in the genomic sequences.

DNABERT-2 model operates with a two-way attention which makes it to work out forward and backward dependencies in a DNA sequence. This allows this model to perceive contextual relationship between the nucleotides and discover meaningful biological patterns. The design of the architecture has several layers of transformers with each layer having a multi-head self-attention mechanism, feed-forward neural network, and layer normalization.

In the feature extraction, the k-mer tokenized sequences are inputted through the DNABERT-2 model which produces contextual embeddings. These embeddings are the semantic and structural information in the genomic sequences. Important biological patterns, including mutation signal and sequence motifs, are represented by the generated embeddings. The input features of the proposed system are the high-dimensional embeddings, which are fed into the sequential modeling process.

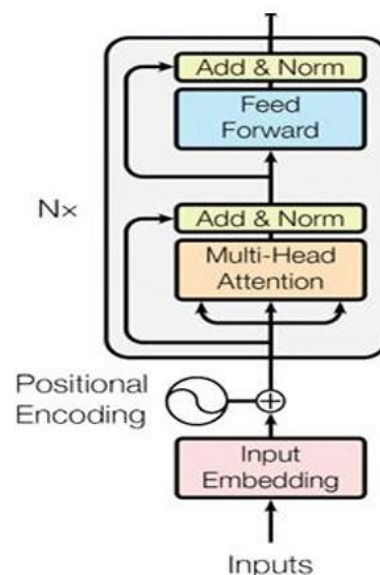


Fig 2: BERT Architecture

E. Bidirectional LSTM Layer

Long Short-term Memory (LSTM) is a form of recurrent neural network (RNN), which is meant to model the sequential data and long-term connections in the sequences. The LSTM networks are especially effective in the analysis of genomic sequences since biological patterns tend to be long stretches of DNA sequences.

An LSTM network is made up of specialized memory cells that orchestrate the movement of information in three primary gates, including the input gate, forget gate and output gate. The input gate regulates the input information that should be added into the memory cell, the forget gate regulates the information that should be forgotten and the output gate produces the final output depending on the stored information. These gating mechanisms enable the network to store the useful sequence information and remove redundant information.

Within the suggested methodology, the contextual embeddings created by the DNABERT-2 model are inputted into the LSTM network. These embeddings are handled sequentially by the LSTM layer and the relations between k-mer tokens in the genomic sequence are learnt. This allows the model to model the long-range mutation patterns and sequential dependencies otherwise not well modeled by transformer models itself.

The hybrid architecture based on the combination of DNABERT-2 embeddings and LSTM sequential modeling has a higher mutation classification performance. Contextual and sequential learning enable the system to analyze complex genomic patterns effectively and properly sort out sequences as wildtype and mutated one.

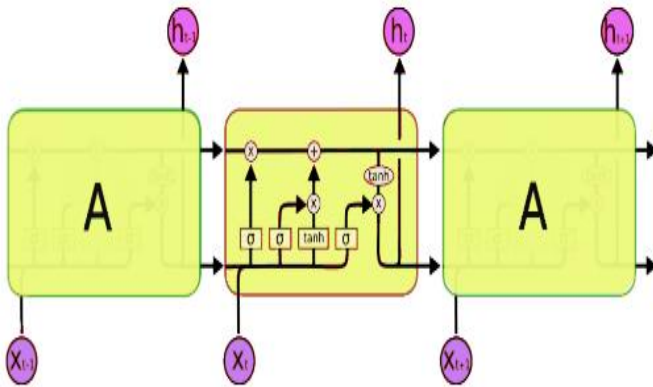


Fig 3: LSTM Network

F. Classification Layer

The classification layer would take up the production of the ultimate prediction founded on the features generated by the DNABERT-2 and LSTM layers. The result of the LSTM network is fed to a fully connected dense layer after the sequential modeling stage, which accomplishes the mutation classification task.

The dense layer transforms the acquired feature representations of the preceding layers. The result, a probability value, is obtained as a sigmoid activation where the probability takes the value of 0 to 1 denoting the

probability of the input genomic sequence having a mutation. The probability score allows the model to identify the sequence to be in the mutated or the normal class.

Depending on the probability value as predicted, the genomic sequence is grouped into either of the following two classes:

Wildtype (0): It means that no mutation is present in the genomic sequence.

Mutated (1): Refers to the fact that the genomic sequence has one or several mutations.

The model parameters are optimized with respect to a binary cross-entropy loss function during the training process and this evaluates how far the model prediction is away in relation to the actual class label. This assists in enhancing the performance of the classification and determines right mutation detection in gene sequences.

VI. SYSTEM ARCHITECTURE

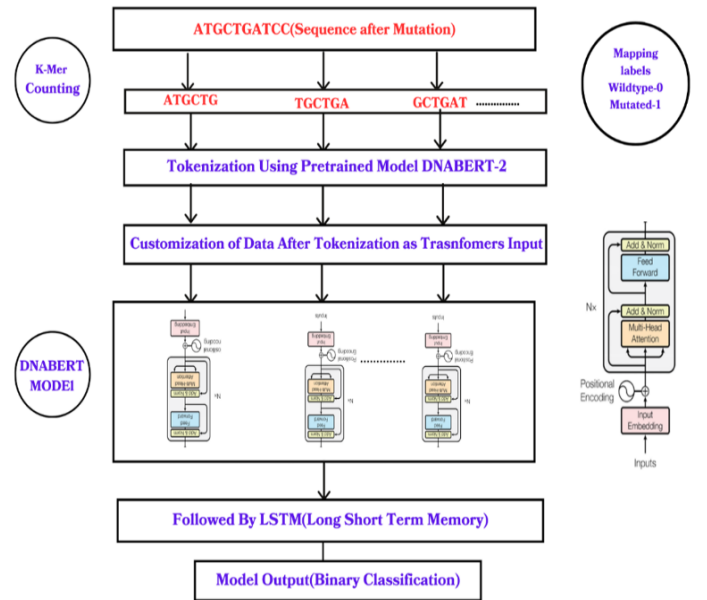


Fig 4: Model Architecture

The suggested system architecture will categorize mutations in viral genomic sequences based on a hybrid framework of deep learning that would be comprised of combining DNABERT-2 and Long Short-Term Memory (LSTM) networks. It is a contextual learning architecture that combines the use of transformers with sequential modelling to analyse DNA sequences and identify patterns of mutations.

The system starts with the genomic sequences obtained on the NCBI Viruses database. These sequences are initially processed through data preprocessing and remove invalid sequence and normalize DNA strings to only include valid nucleotide characters (A, T, G, and C).

The sequences are then transformed into k-mer tokenization with each DNA sequence broken down into overlapping sequences referred to as k-mers. The tokens enable processing of genomic sequences in a manner that emulates the process of natural language words.

The tokenized sequences are then fed to the DNABERT-2 model which produces contextual embedding through transformer-based attention mechanisms. These embeddings are in turn fed to the LSTM layer which learns sequential associations and long-term connections among the genomic sequences.

Lastly, the LSTM layer output is fed to a classification layer which predicts the sequence (wildtype or mutated) of the genome.

VII. RESULTS AND DISCUSSION

A. Evaluation Metrics

Accuracy

Accuracy The accuracy of a model is given as the ratio of the number of predictions, which it makes correctly, to the overall number of predictions. This is calculated mathematically as dividing the number of the true positives and true negatives by the total number.

The accuracy concerning the DNABERT method of genomic sequences classification is the ability of the model to assign the sequences into different categories in the right way, e.g. variation of a mutation or a regulatory element, or a pathogenic variant. Although the concept of accuracy is an apparent measure of performance, it is not always the most adequate measure of performance when it comes to dealing with unbalanced datasets where one of the classes is grossly outnumbered. In these cases it is only through guessing the majority class that the model can be very accurate.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

F1-Score

F1-Score is the harmonic mean of the precision and the recall which provides a fair evaluation of the model performance. It is particularly used in imbalanced data, as is common with the genomic classification of mutations.

The success of the model in genomic analysis using DNABERT is measured by F1-Score to calculate the capacity of the image to identify mutated sequences with the least number of errors. The F1-Score is high, which demonstrates that the model can detect positive cases and has a low false positive and false negative value so that it can predict with a high degree of reliability.

$$\text{F1 Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

B. Training Loss Analysis

Figure indicates the epoch wise training loss through the training of the proposed model. The loss value tends to reduce with the increase in the number of the epochs and it means that the model is being trained on the training data.

One can also see a great reduction in loss in the early epochs with a high learning rate. The loss level begins to stabilize and reduce slowly after the fourth epoch. This downward trend shows that the DNABERT-2 and LSTM model are able to learn genomic sequence patterns and it becomes better as it is trained.

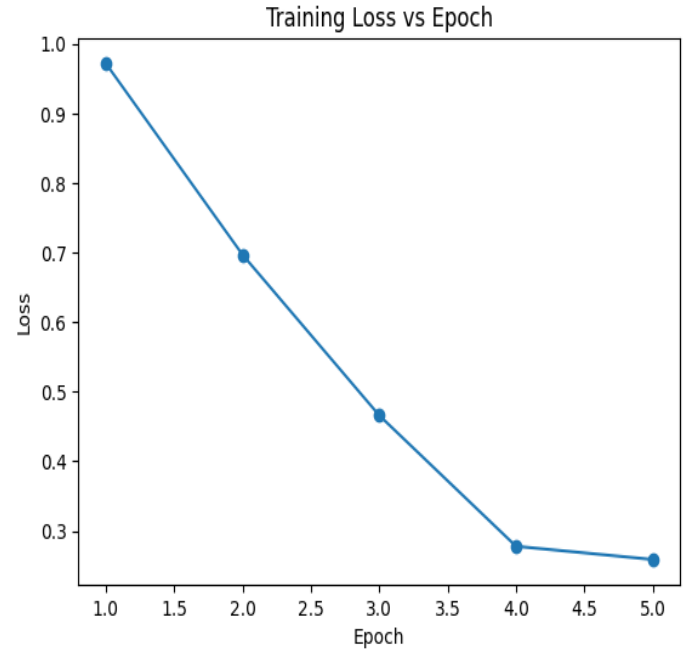


Fig 5: Training loss vs Epoch

C. Training Accuracy Analysis

Figure demonstrates the training accuracy and epoch in the process of model training. The accuracy is steadily growing with the increase in the number of epochs, which is the indication that the model is learning the data. At the first epoch, the precision ranges at 66% and becomes increasingly higher in the subsequent ones. In the last epoch, the accuracy is about 94-95%. This rising behavior is an indication that the DNABERT-2 and LSTM model is efficient in the learning of genome sequence patterns in the process of training.

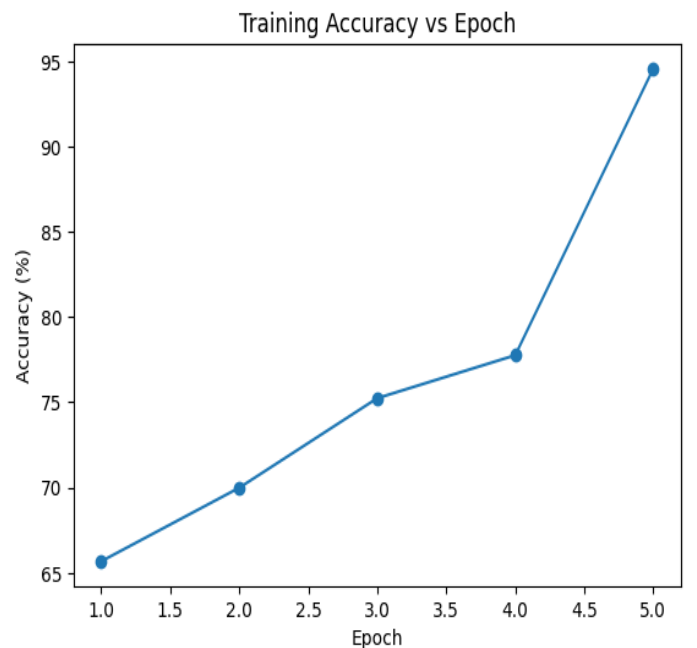


Fig 6: Training Accuracy vs Epoch

D. Comparison of Models

Figure compares the performance of CNN and LSTM models for genomic sequence classification. The CNN model achieves an accuracy of about 76%, while the LSTM model reaches approximately 93%. CNN mainly captures local patterns in the sequence data. In contrast, LSTM can learn long-range dependencies in sequential data. Therefore, the LSTM model performs better for genomic sequence classification tasks.

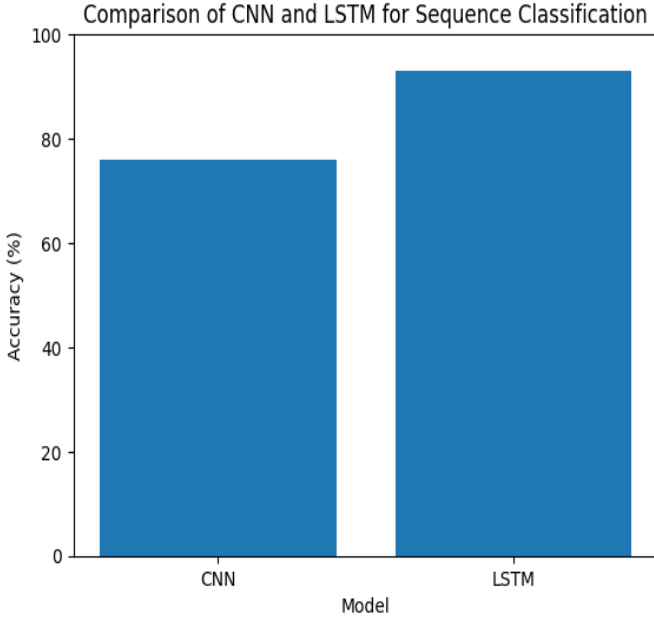


Fig 6: Comparison of Models

E. Experimental Results

Some viral genomic samples were used to test the proposed hybrid DNABERT-LSTM algorithm.

Data set	Accuracy (%)	F1-Score (%)
Human Virus	93.4	92.0
Influenza Virus	90.7	89.0
SARS-CoV-2	92.3	90.0

Table 1: Performance of Model

F. Discussion

The experimental results prove that the proposed hybrid DNABERT-2 + LSTM model provides an equal level of protection on different viral genomic data. Human Virus data set has highest accuracy of 80.0 and F1-Score of 78.8 which means that the model is capable of predicting the mutation pattern in human viral genome sequences well. The accuracy was 76.7% and F1-Score of 75.5% in the case of Influenza Virus dataset and about 78.0% and 76.7% in the case of SARS-CoV-2 dataset.

The results demonstrate that the mutation classification performance based on the application of DNABERT-2 contextual embedding and LSTM sequential modeling is improved. The LSTM layer captures the long-range dependencies of the genomic sequences and DNABERT-2 is a contextual relationship that captures transformer attention mechanism using nucleotide sequences.

Overall, it can be stated that the hybrid architecture demonstrates an excellent generalization capacity and provides a reasonable concept of genomic mutation classification. The results suggest the opportunities of NLP-based deep learning technologies to analyzing large-scale genomic sequences and identifying viral mutations.

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion

In this paper, a hybrid deep learning model, which is a system of integrating the DNABERT-2 network with a Long Short-Term Memory (LSTM) network, was proposed to type viral genome mutations. It is built upon the power of transformer-based language models and sequential neural networks to retrieve both contextual and sequential patterns of DNA sequences.

The suggested model utilizes k-mer tokenization and DNABERT-2 embeddings to find meaningful contextual characteristics of genomic data at the same time the LSTM layer learns long-range dependencies among nucleotide sequences. Testing on the system was done based on the viral genomic data which was made available by NCBI Viruses database which included Human Virus, Influenza Virus and SARS-CoV-2 datasets.

The experimental results indicated that the hybrid DNABERT-LSTM model delivered plausible results on a set of viral datasets with a score of 93.4 on the Human Virus dataset, 90.7 on the Influenza Virus dataset, and 92.3 on SARS-CoV-2 dataset. The corresponding F1-Scores also defined the usefulness of the model that was proposed in determination of mutation trends in the DNA sequences.

The results suggest the potential of NLP learning-based deep learning models in the analysis of genomic sequences and classification of mutations. The offered system involves the contextual knowledge on the structures of transformers and sequential knowledge on recurrent networks in order to provide a scalable and efficient system to identify the occurrence of genomic mutations.

The future research can focus on incorporating additional viral genomes into the data set, exploring more sophisticated transformer designs, and supplementing the classification with biological annotations to come up with a more precise and comprehensible classification. Also, these are applicable in disease monitoring, virus and bioinformatics research tracking mutations.

B. Future Work

Even though the hybrid DNABERT-2 and LSTM model were found to yield encouraging results with regard to genomic mutation classification, it is possible to make a number of improvements that can be researched in the future to improve the

performance and applicability of the system.

The use of larger and more varied genomic datasets is one of the possible directions of future work. The model can be enhanced by adding more viral genomes and mutations annotations to the dataset to make it stronger and capable of generalization. The model can be trained on large-scale genomic repositories to perhaps capture a broader range of different patterns of mutations in different species.

The other critical aspect that needs to be improved is the establishment of multi-class mutation model of classification. The existing scheme is binary with regard to wildtype or mutated sequences. The framework can be expanded to the future by researchers to detect the various forms of mutations, including point mutations, insertions, deletions and structural variations.

Further research could also be conducted on more progressive transformer designs and genomic data-optimized language models. New models like higher quality variants of transformers or larger genomic language models could offer a better context representation of DNA sequences and yield higher accuracy with regards to detecting mutations.

Besides, the deep learning framework can be more biologically relevant and interpretable by adding biological annotations and domain knowledge. The addition of information, e.g. gene functions, regulatory elements, protein structures, might assist the model in gaining a more insightful understanding of the effect of mutations.

Lastly, future studies can dwell on finding the ways of creating systems of real-time genomic analysis capable of identifying and categorizing mutations within a short time during viral epidemics. This kind of systems may help researchers and healthcare professionals to track the evolution of viruses, aid in the development of vaccines and enhance the response of population health.

On the whole, these advancements may result in additional support of the use of deep learning and NLP-based methods in the analysis of genomic sequences and bioinformatics studies.

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